SIMATS ENGINEERING

LAB EXPERIMENTS

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

COURSE CODE: CSA05

EXP 6 : SELECT with various clause – BETWEEN, IN, Aggregate function

```
mysql> create database labexp6;
Query OK, 1 row affected (0.01 sec)

mysql> use labexp6;
Database changed
mysql> create table courses(Coursecode VARCHAR(10) PRIMARY KEY,coursename varchar(45)
,CourseDesc VARCHAR(100));
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE Students (StudentID INT PRIMARY KEY,StudentName VARCHAR(50),Cours
ecode VARCHAR(10),Marks INT,FOREIGN KEY (Coursecode) REFERENCES Courses(Coursecode));
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> INSERT INTO Courses VALUES('C001', 'Statistics', 'Statistical Analysis');
Query OK, 1 row affected (0.06 sec)

mysql> INSERT INTO Courses VALUES('C002', 'Programming', 'Computer Programming');
Query OK, 1 row affected (0.06 sec)

mysql> INSERT INTO Courses VALUES('C003', 'Mathematics', 'Mathematical Concepts');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Courses VALUES('C004', 'Database', 'Database Management');
Query OK, 1 row affected (0.05 sec)

mysql> select * from courses;
+------------+-------------+----------------------+
| Coursecode | coursename  | CourseDesc           |
+------------+-------------+----------------------+
| C001       | Statistics  | Statistical Analysis |
| C002       | Programming | Computer Programming |
| C003       | Mathematics | Mathematical Concepts |
| C004       | Database    | Database Management  |
+------------+-------------+----------------------+
4 rows in set (0.00 sec)

mysql> INSERT INTO Students VALUES
    -> (101, 'Arun', 'C001', 75),
    -> (102, 'Bala', 'C002', 85),
    -> (103, 'Charan', 'C002', 90),
    -> (104, 'Deepak', 'C003', 65),
    -> (105, 'Eswar', 'C004', 80),
    -> (106, 'Farhan', 'C001', 55),
    -> (107, 'Gokul', 'C003', 95),
    -> (108, 'Hari', 'C004', 70);
Query OK, 8 rows affected (0.05 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> select * from students;
+-----------+-------------+------------+-------+
| StudentID | StudentName | Coursecode | Marks |
+-----------+-------------+------------+-------+
|       101 | Arun        | C001       |    75 |
|       102 | Bala        | C002       |    85 |
|       103 | Charan      | C002       |    90 |
|       104 | Deepak      | C003       |    65 |
|       105 | Eswar       | C004       |    80 |
|       106 | Farhan      | C001       |    55 |
|       107 | Gokul       | C003       |    95 |
|       108 | Hari        | C004       |    70 |
+-----------+-------------+------------+-------+
8 rows in set (0.00 sec)
```

```
mysql> select * from courses where coursename in ('statistics','programming');
+------------+-------------+----------------------+
| Coursecode | coursename  | CourseDesc           |
+------------+-------------+----------------------+
| C001       | Statistics  | Statistical Analysis |
| C002       | Programming | Computer Programming |
+------------+-------------+----------------------+
2 rows in set (0.05 sec)

mysql> select studentname , coursecode , marks from students where marks between 50 a
nd 80;
+-------------+------------+-------+
| studentname | coursecode | marks |
+-------------+------------+-------+
| Arun        | C001       |    75 |
| Deepak      | C003       |    65 |
| Eswar       | C004       |    80 |
| Farhan      | C001       |    55 |
| Hari        | C004       |    70 |
+-------------+------------+-------+
5 rows in set (0.00 sec)

mysql> select avg(marks) from students where coursecode = 'C002';
+------------+
| avg(marks) |
+------------+
|    87.5000 |
+------------+
1 row in set (0.04 sec)

mysql> select max(marks),min(marks),avg(marks) from students where coursecode = 'C002
';
+------------+------------+------------+
| max(marks) | min(marks) | avg(marks) |
+------------+------------+------------+
|         90 |         85 |    87.5000 |
+------------+------------+------------+
1 row in set (0.05 sec)
```